Chyi Ricketts

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EDUCATION

Imperial College London *09.2025 – 09.2026*
Master of Research in Cancer Informatics London, United Kingdom
Honors: Distinction (Highest Classification)

Duke University & Duke Kunshan University (Dual Degree) *08.2021 – 05.2025*
BS in Genetics and Genomics Durham, North Carolina
Honors: Graduation with Distinction | GPA: 3.7/4.0 Kunshan, Jiangsu, China

LABORATORY SKILLS

Computational and Bioinformatics Tools

- **Python, R / RStudio, SQL, HTML/CSS, JavaScript**
- **Bioinformatics analysis:** Multi-omics integration, Survival analysis (Cox regression), CNV/methylation analysis
- **Drug discovery:** Scoring functions, Binding affinity prediction
- **Workflow & reproducibility**: Git/GitHub, Docker, Nextflow
- **Machine learning**: scikit-learn, PyTorch, TensorFlow
- **Image analysis**: ImageJ/Fiji, Cellpose, Segment Anything
- **Systems**: Linux (command line), HPC (PBS/qsub), AWS

Wet Lab Techniques

- Mammalian and bacterial cell culture
- PCR and RT-qPCR
- Gel Electrophoresis
- Flow Cytometry
- siRNA/RNA knockdown
- CRISPR knock-in tagging
- Confocal & fluorescence microscopy
- Zebrafish embryo microinjection

RELEVANT RESEARCH EXPERIENCE

Boston University *10.2026 – Present*
Zhang Lab Boston, Massachusetts

- Bioinformatics Analyst: H&E Imaging and Spatial Transcriptomics

Imperial College London *09.2025 – 09.2026*
Ballester Lab London, United Kingdom
Project: Multi-Omics Identification of a Novel Prognostic Biomarker in Glioblastoma Subtypes

- Applied high-throughput multi-omics (RNA-seq, DNA methylation, CNV, RPPA) to TCGA glioblastoma cohorts (n=600+), performing Cox regression and Kaplan-Meier survival analyses to nominate ARSI as a candidate prognostic biomarker
- Identified subtype-specific survival associations in the Mesenchymal GBM subtype and characterized epigenetic upregulation of ARSI via CpG hypomethylation analysis, integrating molecular and clinical covariates

Master's Thesis: Target-Specific Binding Affinity Prediction Scoring Functions in Lead Optimization

- Curated and preprocessed large-scale protein–ligand datasets (~100,000 complexes) with RDKit and ECFP6 Tanimoto FPs
- Built large-scale, reproducible HPC benchmarking pipelines evaluating 100+ protein-ligand models across featurization schemes (AEV-PLIG, ECIF, PLEC) and learning algorithms (GATv2, XGBoost, Random Forest, Support Vector Machine)

Duke Kunshan University *05.2023 – 05.2025*
Huang Lab Kunshan, Jiangsu, China
Thesis: Integrating Segmentation Softwate for Automated Cell Image Analysis in a Web-based Application

- Quantitatively benchmarked generic and fine-tuned Cellpose, Ilastik, and SAM on self-generated RNAi-based HeLa cell imaging data using custom computational metrics, following qualitative screening of candidate segmentation tools
- Developed an interactive visualization tool for post-segmentation image analysis (https://sw-cellvis.onrender.com/)

Grant writing: Authored and submitted a successful grant proposal to the Kunshan Supercomputing Center, resulting in the award of 9 high-performance computing nodes to support image analysis and segmentation software benchmarking

Duke University *04.2022 – 07.2024*
Di Talia Lab & Bagnat Lab Durham, North Carolina
Project: DST SPARK Synchronized Clocks in Zebrafish Patterning

- Trained and benchmarked 10 Cellpose models to analyze oscillating ERK expression in embryonic zebrafish notochords
- Applied Knock-in Tagging methods to examine protein participation in nutrient transport within the interstitial lumen

ADDITIONAL SKILLS

- **Languages**: Native English Speaker, Fluent in Mandarin and Advanced literacy in Traditional and Simplified Chinese